MINISTRY OF SCIENCE AND EDUCATION OF THE REPUBLIC OF AZERBAIJAN
KARABAKH UNIVERSITY Faculty of Mathematics & IT — Computer Engineering

Course: Introduction to Python Programming Laboratory Assignment No: 1 Topic: Conditional Statements and Decision-Making in Python

Student Name: Ibrahim Aliyev Group: KOM26 Instructor: Shamil Humbatov Date: October 13, 2026

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1. OBJECTIVE

The main objectives of this laboratory assignment are:

- To practice and master conditional statements (if, elif, else) in Python.
- To design decision-making logic using mathematical operators and modulo arithmetic.
- To implement repeating user interactions using conditional while loops.
- To gain hands-on experience in processing console inputs and displaying formatted outputs.

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2. TASKS AND RESULTS

Task 1: Body Mass Index (BMI) Calculator

Description: The program prompts the user for their weight in kilograms and height in meters. It then calculates the Body Mass Index using the formula: BMI = weight / (height * height)

Based on the calculated value, the program classifies the user into one of four weight categories:

- Underweight: BMI < 18.5
- Normal Weight: 18.5 <= BMI < 25.0
- Overweight: 25.0 <= BMI < 30.0
- Obese: BMI >= 30.0

Execution Output:

```
Enter weight: 68
Enter height: 1.75
BMI: 22.20408163265306
Category: Normal Weight
Do you want calculate again? (y/n): n
```

Task 2: Chinese Zodiac Calculator

Description: The program determines the user's Chinese Zodiac animal sign based on their birth year. It validates that the year is within the range of 1900 to 2026. The algorithm calculates the index using the modulo operator: index = year % 12

The resulting index maps to a predefined list of 12 zodiac animals: Monkey (0), Rooster (1), Dog (2), Pig (3), Rat (4), Ox (5), Tiger (6), Rabbit (7), Dragon (8), Snake (9), Horse (10), and Sheep (11).

Execution Output:

```
Enter your year: 2008
Your Chinese Zodiac sign is: Rat
Do you want check your Chinese Zodiac sign? (y/n): y
Enter your year: 2027
Please, enter correct year!!
Enter your year: 1899
Please, enter correct year!!
Enter your year:
```

Task 3: Birthday Guessing Game

Description: The program guesses the user's birth day (a number from 1 to 31) without asking for it directly. It displays five predefined sets of numbers based on powers of two (1, 2, 4, 8, 16). The user responds with 1 (Yes) or 0 (No) for each set. The program calculates the birth day by accumulating the base values of all sets where the answer was 1.

Execution Output:

```
Is your birthday in Set1?
 1  3  5  7
 9 11 13 15
17 19 21 23
25 27 29 31
Enter 0 for No and 1 for Yes: 0

Is your birthday in Set2?
 2  3  6  7
10 11 14 15
18 19 22 23
26 27 30 31
Enter 0 for No and 1 for Yes: 1

Is your birthday in Set3?
 4  5  6  7
12 13 14 15
20 21 22 23
28 29 30 31
Enter 0 for No and 1 for Yes: 0

Is your birthday in Set4?
 8  9 10 11
12 13 14 15
24 25 26 27
28 29 30 31
Enter 0 for No and 1 for Yes: 0

Is your birthday in Set5?
16 17 18 19
20 21 22 23
24 25 26 27
28 29 30 31
Enter 0 for No and 1 for Yes: 1

Your birthday is: 18
Do you want to ques your birthday? (y/n): n
```

Your birthday is: 18 Do you want to gues your birthday? (y/n): n

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3. CONCLUSION

In this assignment, I successfully implemented and tested three Python applications utilizing basic decision-making constructs. Through these exercises, I deepened my understanding of conditional logic (if-elif-else), array/list indexing using modulo operators, positional binary arithmetic, and controlling interactive program loops with while statements. All programs performed accurately according to the task requirements.